

Screw-Based Kinematic Control for Multi-Rigid-Body Simulation

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1. Introduction

This document is a basic derivation of kinematic control of multi-rigid-bodies from the perspective of screw theory. In the modern community of robot modelling, screw-based description is popular, as it avoids the singularity of D-H matrix method while being simple enough. To begin with, we give notation in this document.

2. Notations

Let i and j represent different rigid bodies, whose states $\mathbf{H}_i$ and $\mathbf{H}_j$ can be regarded as elements in $SE(3)$. Specifically, the detailed denotation of $\mathbf{H}_i$ satisfy the following equations according to the left-invariant convention:

$$\mathbf{H}_i = \begin{bmatrix} \mathbf{R}_i & \mathbf{p}_i \\ \mathbf{0} & 1 \end{bmatrix} \quad \delta \mathbf{H}_i = \mathbf{H}_i \delta \mathbf{h}_i^\wedge \quad \dot{\mathbf{H}}_i = \mathbf{H}_i \mathbf{v}_i^\wedge \quad (1)$$

where $\mathbf{R}_i$ and $\mathbf{p}_i$ respectively stand for the rotation and translation of rigid body i , $\delta \mathbf{h}_i^\wedge$ and $\mathbf{v}_i^\wedge \in se(3)$ are the Lie algebra elements representing the variation of state and the velocity, whose vectorized forms are given as $\delta \mathbf{h}_i$ and $\mathbf{v}_i$. Since $\delta \mathbf{h}_i$ and $\mathbf{v}_i$ are obtained from different operations, here $\mathbf{q}_i$ is used as an integrated term of $\mathbf{v}_i$.

3. Index-3-Level Kinematic Constraints

The governing equations of multibody system dynamics are differential algebraic equations (DAEs), usually read in following Index-3 form:

$$\begin{cases} \mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{B}(\mathbf{H})^T \boldsymbol{\lambda} = \mathbf{F}_{\text{ext}} \\ \boldsymbol{\Phi}(\mathbf{H}) = \mathbf{0} \end{cases} \quad (2)$$

with $\boldsymbol{\Phi}(\mathbf{H})$ being the constraint equations and $\mathbf{B}(\mathbf{H})$ being the constraint gradients, satisfying

$$D\boldsymbol{\Phi}(\mathbf{H})\delta\mathbf{h}^\wedge = \mathbf{B}(\mathbf{H})\delta\mathbf{h} \quad (3)$$

it is convenient to construct the kinematic control constraints using

$$\mathbf{H}_i = \mathbf{H}_j \mathbf{S}_{ji}(t) \quad (4)$$

where $\mathbf{S}_{ji} \in \text{SE}(3)$ is a screw in the frame of the j the rigid body. To formulate Eq. (4) in vectorized form, the logarithm map is used

$$\boldsymbol{\Phi}(\mathbf{H}) = \log_{\text{SE}(3)} \left(\mathbf{H}_i^{-1} \mathbf{H}_j \mathbf{S}_{ji} \right)^\vee \quad (5)$$

Where $^\vee$ stands for the vectorize transformation of $\text{se}(3)$ elements. To simplify the following derivation process, we denote $\mathbf{Y} = \mathbf{H}_i^{-1} \mathbf{H}_j \mathbf{S}_{ji}$. Assume the states of rigid bodies are disturbed such that

$$\mathbf{H}'_i = \mathbf{H}_i \exp_{\text{SE}(3)}(\delta\mathbf{h}_i^\wedge) \quad \mathbf{H}'_j = \mathbf{H}_j \exp_{\text{SE}(3)}(\delta\mathbf{h}_j^\wedge) \quad (6)$$

If $\mathbf{Y}$ is disturbed by the i th rigid body, then

$$\mathbf{Y}'_i = \mathbf{Y} \mathbf{Y}^{-1} \exp_{\text{SE}(3)}(-\delta\mathbf{h}_i^\wedge) \mathbf{Y} = \mathbf{Y} \exp_{\text{SE}(3)} \left[-(\mathbf{Ad}_{\mathbf{Y}^{-1}} \delta\mathbf{h}_i)^\wedge \right] \quad (7)$$

Similarly, if $\mathbf{Y}$ is disturbed by the j th rigid body, then

$$\mathbf{Y}'_j = \mathbf{Y} \mathbf{S}_{ji}^{-1} \exp_{\text{SE}(3)}(\delta\mathbf{h}_j^\wedge) \mathbf{S}_{ji} = \mathbf{Y} \exp_{\text{SE}(3)} \left[(\mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \delta\mathbf{h}_j)^\wedge \right] \quad (8)$$

Notice that, Eq. (5) actually represent the Lie algebraic element of $\mathbf{Y}$. To find out the influence of disturbance on rigid body states in Eq. (5), we have

$$\exp_{\text{SE}(3)} \left[(\mathbf{y} + \delta\mathbf{y})^\wedge \right] = \exp_{\text{SE}(3)}(\mathbf{y}^\wedge) \exp_{\text{SE}(3)} \left[(\mathbf{T}(\mathbf{Y}) \delta\mathbf{y})^\wedge \right] \quad (9)$$

where $\mathbf{T}(\mathbf{Y})$ stands for the tangent map of exponential map for $\text{SE}(3)$. In this way, the disturbance $\delta\mathbf{y}$ of Eq. (7) satisfies

$$\delta\mathbf{y}_i^\wedge = - \left[\mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{Y}^{-1}} \delta\mathbf{h}_i \right]^\wedge \rightarrow \delta\mathbf{y}_i = -\mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{Y}^{-1}} \delta\mathbf{h}_i \quad (10)$$

Also,

$$\delta\mathbf{y}_j = \mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \delta\mathbf{h}_j \quad (11)$$

By use Eq. (10) and Eq. (15), it can be found that

$$\frac{\partial \Phi}{\partial \mathbf{h}_i} = -\mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{Y}^{-1}} \quad \frac{\partial \Phi}{\partial \mathbf{h}_j} = \mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \quad (12)$$

Then

$$\mathbf{B}(\mathbf{H}) = \begin{bmatrix} -\mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{Y}^{-1}} & \mathbf{T}(\mathbf{Y})^{-1} \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \end{bmatrix} \quad (13)$$

4. Index-2-Level Kinematic Constraints

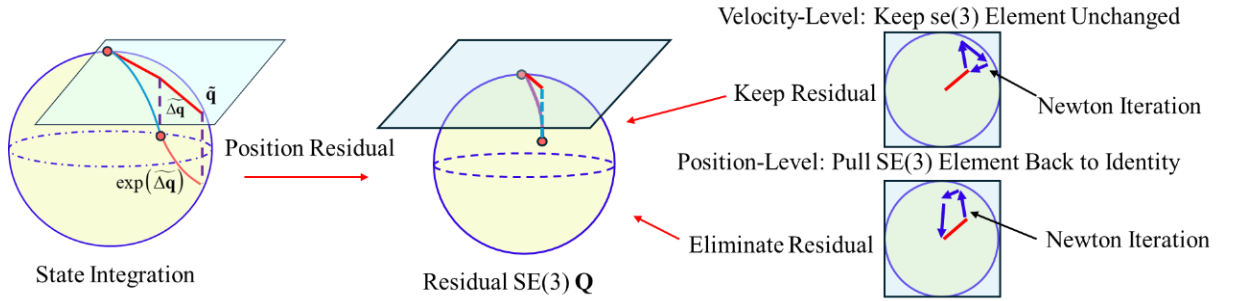


Fig 1. Iteration Result by Different Constraint Equations

It is obvious that the state of rigid bodies does not explicitly appear in Eq. (2). A possible solution is to directly construct the kinematic constraint equation on the velocity level. With Eq. (4), we have

$$\dot{\mathbf{H}}_i = \dot{\mathbf{H}}_j \mathbf{S}_{ji}(t) + \mathbf{H}_j \dot{\mathbf{S}}_{ji}(t) \quad (14)$$

$$\mathbf{H}_i \mathbf{v}_i^\wedge = \mathbf{H}_j \mathbf{v}_j^\wedge \mathbf{S}_{ji} + \mathbf{H}_j \mathbf{S}_{ji} \dot{\mathbf{s}}^\wedge \quad (15)$$

Here the adjoint operation can be used again, so that

$$\mathbf{v}_i = \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \mathbf{v}_j + \dot{\mathbf{s}} \quad (16)$$

Then we have the velocity-level constraint

$$\Phi(\mathbf{v}) = \mathbf{v}_i - \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \mathbf{v}_j - \dot{\mathbf{s}} \quad (17)$$

Obviously in this way, the constraint equation becomes linear. It is theoretically simple and elegant. By making comparison with Eq. (13), the Jacobian matrix of Eq. (17) is a

degenerated form of Eq. (13) at the identity element of $\mathbf{Y}$. Eq. (17) means that at each integration step, the position-level violation is ignored, i.e. $\mathbf{Y} \equiv \mathbf{I}$. Moreover, tangent map is not appeared in Eq. (17), which means that velocity-level constraint cannot modify position-level violation. As shown in Fig. 1, when an implicit integration step is conducted, Eq. (13) and Eq. (17) lead to different iteration directions. As a result, Eq. (17) often leads to position-level violation once the step size is large.

5. Modified Index-2-Level Kinematic Constraints

The operation in Sect. 4 is logically correct. If we have an ideal integrator that has an infinitely small integration step size, Eq. (17) can successfully maintain the position-level constraints. However, in real cases, the integrators must have a finite integration step size, which results in the position-level violation. But fortunately, we only integrate variables in linear spaces. Still, we focus on the position-level constraint equations and we give a finite disturbance

$$\mathbf{Y}' = \mathbf{Y} \exp_{\text{SE}(3)}(\delta \mathbf{y}^\wedge) \quad (18)$$

The time derivative of Eq. (18) reads

$$\dot{\mathbf{Y}}' = \mathbf{Y} \frac{d \exp_{\text{SE}(3)}(\delta \mathbf{y}^\wedge)}{dt} \quad (19)$$

The actual velocity of $\mathbf{Y}'$ satisfies

$$\mathbf{v}_{\mathbf{Y}} = \mathbf{T}(\mathbf{Y}) \delta \dot{\mathbf{y}} \quad (20)$$

It is easy to obtain

$$\mathbf{v}_{\mathbf{Y}}^\wedge = -\mathbf{Y}^{-1} \mathbf{v}_i^\wedge \mathbf{Y} + \mathbf{S}_{ji}^{-1} \mathbf{v}_j^\wedge \mathbf{S}_{ji} + \dot{\mathbf{s}}_{ji}^\wedge \rightarrow \mathbf{v}_{\mathbf{Y}} = -\mathbf{Ad}_{\mathbf{Y}^{-1}} \mathbf{v}_i + \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \mathbf{v}_j + \dot{\mathbf{s}}_{ji} \quad (21)$$

The velocity-level disturbance

$$\delta \dot{\mathbf{y}} = \mathbf{T}(\mathbf{Y})^{-1} \left(-\mathbf{Ad}_{\mathbf{Y}^{-1}} \mathbf{v}_i + \mathbf{Ad}_{\mathbf{S}_{ji}^{-1}} \mathbf{v}_j + \dot{\mathbf{s}}_{ji} \right) \quad (22)$$

In this way, we succeed in linearizing the constraint equation while maintaining correct iteration direction. The key operation is to consider disturbance instead of infinitesimal small differential operation.